Q2 Let $a_1, a_2, \dots, a_{2^n}$ be the players.

There are $\binom{2^n}{2}$ pairs of players. (or we say we count the size of $\underline{\underline{\{\{a_i, a_j\} : i \neq j\}}}$)

$$\binom{2^n}{2} = \frac{2^n(2^n-1)}{2} = 2^{n-1}(2^n-1) =: N$$

Let

$\Omega := \{b_1, \dots, b_N\}$ be the set of all ~~such~~ pairs of players.

Let p_k^ℓ = the probability that the pair b_ℓ meets in the kth round.

$\Rightarrow p_k^\ell = \dfrac{|E_k^\ell|}{|X|}$ where $E_k^\ell := \{$ Tournaments where b_ℓ meets in the kth round $\}$

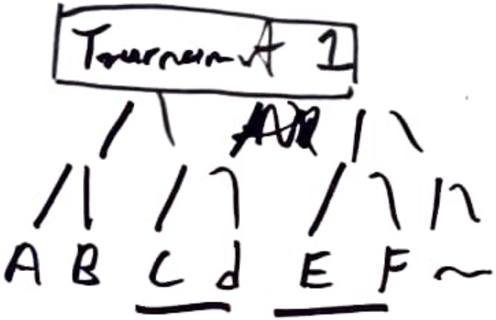
$X := \{$ All tournaments $\}$

Double counting: Total numbers of matches in the kth round throughout all tournaments.

I just write $Total_k$

~~Overlap~~ Multiple instances of the same elements is allowed (multiset)

Exp.



The match A ~~vit~~ vs B in Tournament 1 is not the same as Tournament 2. So we will count A vs B twice.

① $Total_k$ = Total number of tournaments

$\times$

Total number of matches in the kth round in a tournament $=: m_k$

$= |X| \cdot m_k$

(2) $\boxed{\begin{array}{l} \text{Total}_k = \text{Total number of pairs} \\ \qquad\qquad \times \\ \qquad \text{Total number of Tournaments where } b_\ell \\ \qquad \text{meets in the } k\text{th round.} \end{array}}$

~~Reasoning: Fix a Choose~~

$$\therefore \text{Total}_k = |\Omega| \cdot |E_k^\ell|$$

Reasoning: Choose a pair b_ℓ of ~~p~~ players. ($|\Omega|$ of them)
To count the number of matches, throughout all tournaments, that contains b_ℓ in the kth round

it is just $|E_k^\ell|$.

By double counting,

$$|X| \cdot m_k = |\Omega| \cdot |E_k^l|$$

$$\Rightarrow p_k^l = \frac{|E_k^l|}{|X|} = \frac{m_k}{|\Omega|}$$

In fact I can drop l since every pair would have the same probability.

Recall that m_k = # ~~[illegible]~~ matches in the kth round in a tournament.

(i) 1st round $\Rightarrow k=1$

$\Rightarrow$ There are 2^{n-1} matches in the 1st round

$$\Rightarrow p_1 = \frac{2^{n-1}}{|\Omega|} = \frac{2^{n-1}}{2^{n-1}(2^n-1)} = \frac{1}{2^n-1}$$

(ii) Final round $\Rightarrow$ Only one match

$\therefore\ m_{last} = 1$

$\therefore\ p_{last} = \dfrac{1}{|\Omega|} = \dfrac{1}{2^{n-1}(2^n-1)}$

(iii) the kth round $\Rightarrow$ $m_k = 2^{n-k}$ matches

$\therefore\ p_k = \dfrac{2^{n-k}}{|\Omega|} = \dfrac{2^{n-k}}{2^{n-1}(2^n-1)} = \dfrac{1}{2^{k-1}(2^n-1)}$